

STUDIES ON STEAM STERILIZATION OF SOILS

I. SOME EFFECTS ON PHYSICAL, CHEMICAL, AND BIOLOGICAL PROPERTIES¹

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Abstract

Effects of steam sterilization and recontamination with original soil, on certain physical, chemical, and biological properties of four Alberta soils were investigated. The soils were Edmonton black loam, Vegreville black fine sandy loam, Gros Ventre brown loam, and Fallis gray silt loam. Certain physical properties were affected by steam sterilization as follows: (1) capillary rise of moisture was greatly retarded in all four soils following sterilization; (2) the percentages of the different separates as determined by ordinary mechanical analysis were not changed significantly by sterilization; (3) water-holding capacity and moisture content at the 'sticky point' were slightly reduced, and, in three of the soils, shrinkage was slightly increased by sterilization, but these effects were of uncertain significance; (4) the pH values of these soils were not changed appreciably by sterilization. The following chemical changes were produced by steam sterilization: (1) water-soluble phosphate was generally greatly increased and water soluble sulphate somewhat increased by sterilization. Easily soluble phosphorus (soluble at pH 3) was increased by about one-third in the two black soils rich in organic matter, but not in the other soils; (2) ammonia accumulated rapidly in the sterilized recontaminated black and brown soils for four to six weeks, and then returned to normal in 8 to 12 weeks. Ammonia accumulated less rapidly and to a smaller degree in the gray soil; (3) nitrification was suppressed for about 6 to 10 weeks in the sterilized recontaminated soils, but was later more active in these soils than in the unsterilized (especially in the black soils rich in organic matter).

Introduction

The study of some physical, chemical, and biological changes brought about by steam sterilization of soils, as reported in this paper, is part of a broader study of the inter-relationships of soil micro-organisms and plant diseases, and of soil micro-organisms and soil fertility. The study of these complicated inter-relationships may be somewhat simplified by first studying the relationships of a few known organisms on sterilized soil devoid of its enormous population. However, the study of effects of known micro-organisms in sterilized soil necessitates a study of effects produced by sterilization, because the sterilized soil may be quite different from the original soil, physically and chemically, and the results obtained may be misleading if the changes due to sterilization are not measured.

In practice, mainly in greenhouses, soils are partially sterilized to get rid of weeds, insects, and other undesirable or disease producing organisms. Complete sterilization is neither strived for nor desirable since there are many organisms in the soil that are beneficial to plant growth because they are

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responsible for the decomposition of organic matter and the elaboration of plant nutrients.

Literature dealing with the changes effected when soils are completely sterilized by steam is less abundant than that dealing with partial sterilization. Effects of various heat and chemical treatments in partially sterilized soils have been studied by many workers in England and the United States, and their findings would apply, in varying degrees, to the Alberta soils used in these investigations. However, different soils respond differently to such treatments, and, furthermore, the Alberta soils were sterilized completely rather than only partially.

Darbishire and Russell (7) noted that in steam sterilized soils heated to 120° C. the absorption of oxygen was greatly diminished, but that in partially sterilized soils, heated to 95° C., the absorption was considerably increased. They found that the crop could get from heated soils three times as much nitrogen and twice as much phosphorus and potash as from the unheated soils. They believed that the increased availability of the plant nutrients was connected with the modification of microbial flora, although they recognized that when soils were heated or sterilized by steam, a certain amount of decomposition took place.

Russell and Hutchinson (16) found that heating the soil by steam to 125° C. killed all the organisms present, and caused a production of 5 p.p.m. of ammonia, but after that first production no further change took place. Heating by steam to 98° C., or treating with toluene, increased ammonification greatly for a time, and they concluded that these treatments destroy organisms that are inimical to ammonifying bacteria. The number of bacteria increased greatly following partial sterilization, and this they attributed to the absence of protozoa. They found that heating to 98° C. not only destroyed the nitrifiers, but made the soil temporarily unfit for their development.

Lyon and Bizzell (11) observed that steamed soil, on standing up to three months without plants, steadily decreased in water soluble organic matter, ammonia, and other soluble nitrogenous matter. They found that nitrates remained low, up to 14 weeks in steamed and recontaminated soils, but that ammonia increased. Baldwin (1) reports a consistent increase in acid soluble phosphorus after sterilization. Pickering (14) and Newhall (13) found a lower water-holding capacity in steam sterilized soil.

Practical sterilization is discussed by Bewley (2), Senner (18), and Newhall (13). Newhall concludes that steam sterilization leaves no toxic residues or water-logging and leaves the soil in good tilth.

Methods

The investigation of changes effected by steam sterilization of Alberta soils and subsequent microbial activities following recontamination was divided into two parts: first, the physical and second the chemical-biological changes. Four Alberta soils, Edmonton black loam, Vegreville black fine sandy loam,

Gros Ventre brown loam, and Fallis gray silt loam, were studied. The first two are high in organic matter and the last, very low.

In the study of the physical properties of these soils, the non-sterilized were compared with the sterilized with respect to mechanical analysis, water-holding capacity, shrinkage, sticky point, and capillary power. Since soil moisture relations affect greatly the ultimate productive power of the soil, certain differences in sterilized soils may be due to purely physical factors.

The chemical-biological studies included determinations of acidity, ammonia, nitrates, sulphates, easily soluble phosphorus, and ammonia production by pure cultures.

In the study of physical and biochemical properties, the non-sterilized soils were compared with the sterilized, and with the sterilized-recontaminated also, in the case of the biochemical properties.

For the study of physical properties the soils were sterilized in 1-gal. crocks, and for the study of biochemical properties the soils were placed in Erlenmeyer flasks and incubation was carried on in a dark control chamber at room temperature (23° C. to 27° C.).

Two nitrification experiments were carried out, the first for a period of 12 weeks and the second for 39 weeks. Nitrates were determined frequently throughout these periods, and one flask or tumbler was taken for each determination. Similar plans were followed in the case of the ammonification and sulphofication experiments. Easily soluble phosphorus was determined on the same sample as ammonia. The experiment on ammonia production by pure cultures of *Ophiobolus* and two other fungi, one antagonistic and the other not antagonistic to *Ophiobolus*, was carried out in Erlenmeyer flasks of sterilized Edmonton black soil inoculated with these organisms. The results of these experiments with pure cultures will be published in a subsequent paper together with results of other similar experiments.

Representative samples of the surface six inches of the four soils were obtained, air-dried, put through a 2 mm. sieve, and well mixed. The soils were brought to optimum moisture and allowed to stand for two days or less before sterilizing. Table I shows the optimum moisture used for each soil and a few other physical characteristics of the four soils.

Sterilization was done by steaming the soil at 15 to 17½ lb. pressure (approximately 120° C.) for 45 min. and again next day at the same pressure for 30 min.

After sterilization the soils were cooled and half of the sterile flasks were recontaminated for nitrification, ammonification, and sulphofication experiments.

In all cases where the soil was incubated, the flasks were taken out of the chamber every week or 10 days and brought up to optimum moisture with sterile water. By means of a stirring rod that had been placed in each flask through the cotton plug, the soils were cultivated after being watered.

TABLE I
PHYSICAL CHARACTERS OF THE FOUR ALBERTA SOILS

	Treatment	Hygroscopic moisture, %	Moisture-holding capacity, %	Optimum moisture, %	Real specific gravity	Sticky point moisture, %
Edmonton black park (loam to silt loam)	Non-sterilized		76		2.34	60
	Sterilized	5.96	72	36	2.40	58
Vegreville black park (sandy loam to fine sandy loam)	Non-sterilized		71		2.51	52
	Sterilized	4.52	71	30	2.54	48
Gros Ventre brown prairie (loam)	Non-sterilized		75		2.32	50
	Sterilized	2.13	74	32	2.33	46
Fallis gray wooded (silt loam)	Non-sterilized		59.5		2.54	33
	Sterilized	1.22	58.5	19	2.51	33

For mechanical analysis, Bouyoucos' hydrometer method (3) was adopted. Moist samples were used rather than dry ones because it was thought that following sterilization drying might affect the soil colloidal content and thereby alter changes induced by steam sterilization. The non-sterilized samples had the same amount of moisture added as the sterilized ones and were kept moist for the same length of time prior to mechanical analysis. Since it is more difficult to weigh accurately moist samples, owing to loss of moisture, a greater analytical error had to be contended with. Otherwise, in the main, the procedure recommended by Bouyoucos was followed. The samples were dispersed by adding water, treating with solutions of sodium oxalate and sodium silicate, and mixing in a mechanical stirrer.

Some seven determinations were made also by the pipette method for comparison with the hydrometer method.

The real specific gravity of these soils (specific gravity of soil particles) was calculated from water displaced in a volumetric flask by 10 gm. of oven-dry soil.

Water-holding capacity was found by the funnel method, using 50 gm. of soil. Readings were taken when drainage stopped, and in certain determinations readings were made at 15 min., one hour, and two hours.

The 'sticky point' moisture content of the sterilized and non-sterilized soils was determined by a somewhat arbitrary method. Air-dry 50 gm. samples were moistened with a measured amount of water and the moisture was worked into the soil with a spatula on a glass plate until the sticky point was nearly reached. This was judged by the appearance of wetness. Then the wet soil was kneaded by hand and a little more moisture worked into it, until the point of maximum stickiness was attained as judged by the adhesion to the palm of the hand and the force required to tear away the soil cake after it was squeezed in the hands.

Shrinkage was determined by a modification of Haines' mercury displacement method (8). The soils were brought to optimum moisture content, and half of each was sterilized as previously described. Then both the non-sterilized and the sterilized soils were air-dried and passed through a 100 mesh sieve. Into 50 gm. of soil on a glass plate a measured amount of water was incorporated by means of a spatula, until the sticky point was nearly reached. It was kneaded, shaped by hand, and moistened, more especially at the ends, to prevent cracking. The wet cylinder was weighed and its volume determined by placing it in a large glass test tube, which was then filled with mercury, and levelled off with a microscope slide. The buoyant effect was overcome by placing a perforated rubber stopper below the surface of the mercury. Weight and volume of the soil cylinder were measured at intervals of drying, first at room temperature, later in an oven at 100° C., and finally after drying for two days at 110° C. All data were reduced to the basis of 1 cc. oven-dry soil. Some difficulty arises in finding the first volume if the soil cylinder is too wet, as the buoyant force of the mercury compresses the soil and distorts the soil cylinder; also, small bubbles often adhere. But with a little drying these difficulties can be overcome and the volume measured accurately.

Capillary rise of water in the sterilized and unsterilized soils was determined by measuring the rise in glass tubes about $1\frac{1}{8}$ in. in diameter. The soils were prepared, sterilized, and air-dried as previously described. The air-dried soil was then put through a 2 mm. sieve, and the tubes, wrapped at the lower end with cheesecloth, were filled uniformly with the soils. The tubes were set vertically in a pan to which water was added and maintained at a constant level. From the time of addition of water, measurements of capillary rise above the water level in the pan were taken at 5 min., 15 min., 1 hr., 2, 3, 4, 12, and 24 hr. and thereafter every day for 42 days, and at longer intervals up to 61 days from the beginning.

The pH values were determined by the quinhydrone electrode method. One day after sterilization, moist samples of sterilized and unsterilized soil equivalent to 5 gm. of water-free soil were diluted with 20 cc. of distilled water and pH values read at 1, 5, 15, and 30 min.

Water soluble phosphorus was determined colorimetrically on 1 : 5 water extract of soil, by the Parker and Fudge method (15). Moist samples equivalent to 50 gm. water-free soil were shaken for one hour and filtered through Berkefeld filters. Then 25 or 50 cc. portions of filtrate were evaporated with a little magnesium nitrate and ignited to remove organic matter before determining phosphorus. However, in a few cases, for purposes of comparison, the dried filtrates were not ignited.

Easily soluble phosphorus (soluble at pH 3) was determined colorimetrically by Truog's (19) method. Moist samples of soil equivalent to 1 gm. water-free soil were extracted by shaking for one hour with 200 cc. buffered pH 3.0 sulphuric acid solution, adjusted when necessary by the addition of one to seven drops of normal sulphuric acid. The pH values were determined with

the potentiometer before and after extraction. A few soil extracts, after being filtered through paper, were highly coloured and such extracts were cleared by means of activated charcoal. Blue colour was developed with ammonium molybdate-sulphuric acid solution and stannous chloride.

Nitrates were determined colorimetrically on 1 : 5 water extracts of soil by the phenol-disulphonic method as modified by Harper (9).

Ammonia was determined by the McLean and Robinson (12) method of leaching 25 gm. of soil with 500 cc. normal sodium chloride solution, distilling the soil extract with magnesium oxide, and collecting the ammonia in standard acid.

Water soluble sulphates were determined gravimetrically. The 1 : 5 extracts were filtered through Berkefeld filters, evaporated with magnesium nitrate, and ignited to destroy organic matter prior to solution in dilute hydrochloric acid and precipitation as barium sulphate.

Results and Discussion

MECHANICAL ANALYSIS

Mechanical analysis of 25 samples of soil were made and the results of 16 are represented in Fig. 1. Taking the silt particles limits as 0.002 mm. to 0.05 mm. in diameter, the Edmonton black park soil should be classed as loam to silt loam, the Vegreville black park soil as sandy loam (probably fine sandy loam), the Gros Ventre brown prairie soil as loam, and the Fallis gray wooded soil as silt loam. Differences between the sterilized and unsterilized soils were not obtained by the methods of mechanical analysis used in these experiments.

WATER-HOLDING CAPACITY

The funnel method of determining water-holding capacity was used and the results indicate that the sterilized soils have about 1% lower water-holding capacity than the unsterilized, but the differences were not obtained in every case and cannot be taken as definitely significant. The water-holding capacities are shown in Table I. Pickering (14) and Newhall (13) report that sterilized soils have a lower water-holding capacity than unsterilized, but that the former hold water for a longer time.

SHRINKAGE

The results of the shrinkage experiments are represented by Figs. 2 and 3. They show that during the early stages of drying there was a decrease in volume proportional to the amount of water lost. This decrease, represented graphically, is shown by the slant of the curve at an angle of approximately 45° to the abscissa.

There was a sudden change in the direction of the curves in Fallis gray wooded soil from about 0.34 cc. moisture per gram water-free soil up to the last stage of drying; the curves become almost parallel to the base (abscissa).

This stage of drying is accompanied by a loss in moisture that is relatively large in comparison with the loss in volume, and is called 'residual' shrinkage.

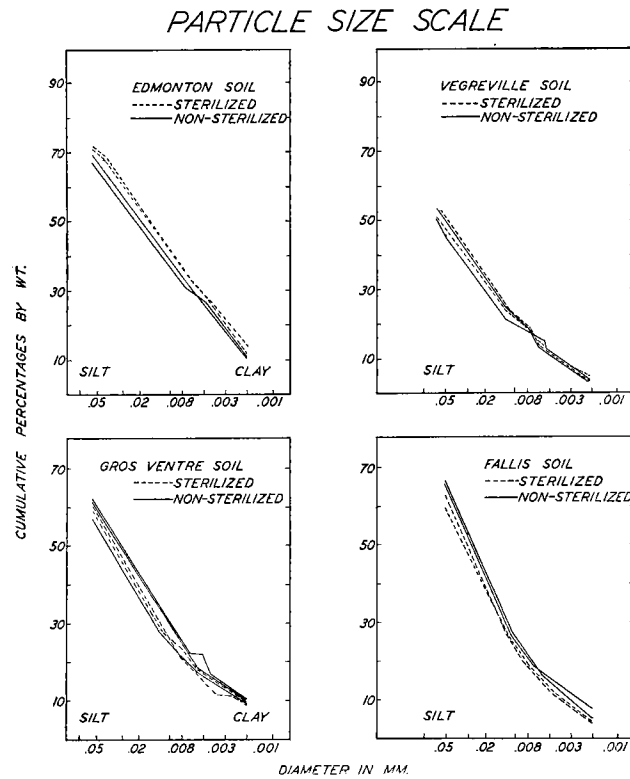


FIG. 1. Mechanical analysis of non-sterilized and steam sterilized Edmonton and Vegreville black park soils, Gros Ventre brown prairie soil, and Fallis gray wooded soil.

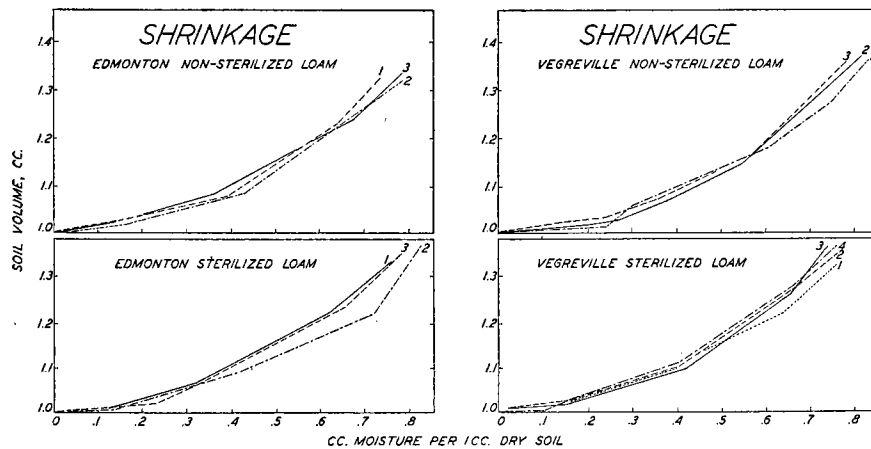


FIG. 2. Shrinkage of non-sterilized and steam sterilized Edmonton and Vegreville black park soils.

This residual shrinkage in the unsterilized Fallis soil amounted to approximately 0.02 cc. decrease in volume for 0.35 cc. of moisture lost, and, in the sterilized, 0.025 cc. for 0.33 cc. moisture lost.

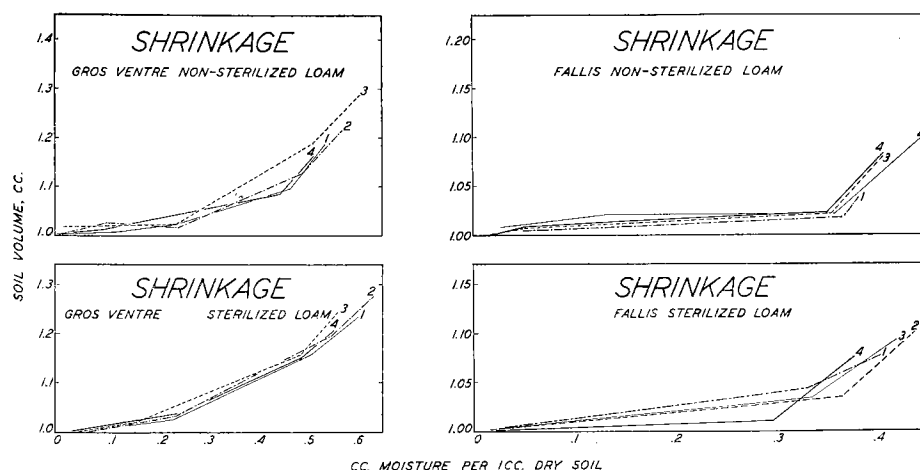


FIG. 3. Shrinkage of non-sterilized and steam sterilized Gros Ventre brown prairie soil and Fallis gray wooded soil.

The break between the 'normal shrinkage' and the 'residual' shrinkage is not so sharp in the Gros Ventre brown prairie soil as might have appeared if a determination had been made at about 0.37 cc. moisture content per cc. of dry soil. The volume of soil at 0.24 cc. of moisture, which is well within the region of 'residual' shrinkage, is, in the unsterilized, approximately 1.02 cc. while in the sterilized at the same moisture it is approximately 1.03 to 1.04 cc. Again this is a small difference, but the shrinkage is greater in the sterilized.

The Vegreville soil shows a gradual change instead of a sharp break. This is indicated by the points plotted between 0.1 cc. and 0.65 cc. of moisture. A slightly greater shrinkage in the case of the sterilized as compared with the unsterilized soil, of about the same magnitude as in the former two soils, was obtained below a moisture content of approximately 0.25 cc. The Edmonton soil shows the same trends as the Vegreville soil.

From Figs. 2 and 3 it appears that 'residual' shrinkage is greatest in Edmonton black soil, followed by Vegreville black, and least in Fallis gray wooded. This indicates that soils high in organic matter have greater 'residual' shrinkage than soils low in this respect, and agrees with the findings of Haines (8) who obtained the highest 'residual' shrinkage, 18.2% volume, on drying peat soil below 50.5% moisture, and only 2.2% in Rothamsted loam below 18.7% moisture. The low 'residual' shrinkage of Fallis gray wooded soil indicates a loss of colloidal material from the soil, probably by leaching, since it is a badly leached soil.

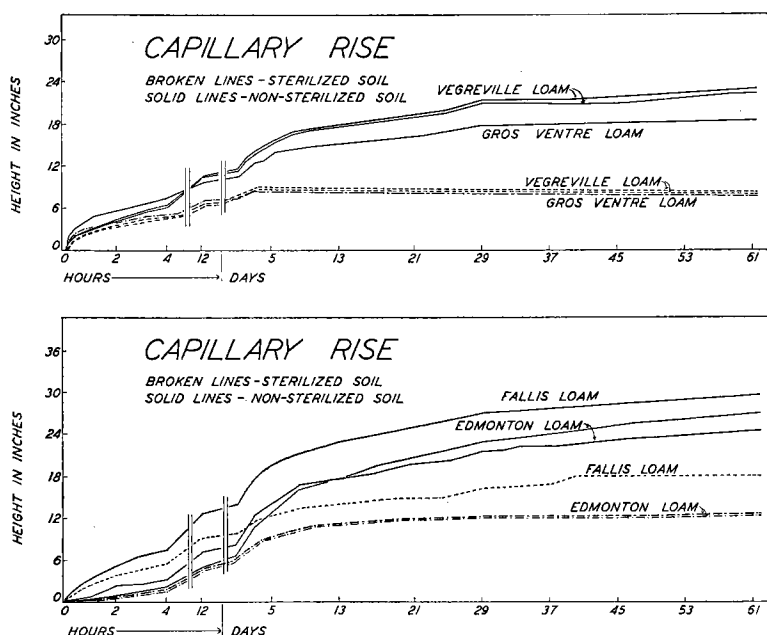
STICKY POINT

The 'sticky point' or point of maximum stickiness appears to be reached at a lower moisture content in sterilized soils than in the unsterilized. It was reached when Edmonton unsterilized soil was brought to 60% moisture (air-dry basis) and sterilized soil was brought to 58%. At a given moisture content, approximately 40% moisture, Edmonton sterilized was more sticky than unsterilized soil. Vegreville, Gros Ventre, and Fallis unsterilized attained sticky points at 52, 50, and 33% of moisture.

On cultivating the soils in the flasks incubated for nitrification, it was observed that those that were kept sterile for a long time became quite sticky upon repeated addition of water to maintain weight for optimum moisture, while the unsterilized and the sterilized recontaminated, with the same moisture content, were not inclined to become sticky.

CAPILLARY RISE

Figs. 4 and 5 show the results of the capillary rise experiment. Consistent and definite differences were found. In all cases the capillary rise after the first 12 hr. was greater in the unsterilized soils. Differences became



FIGS. 4 and 5. Capillary rise in non-sterilized and steam sterilized Edmonton and Vegreville black park soils, Gros Ventre brown prairie soil, and Fallis gray wooded soil.

apparent even after 15 min. in Fallis and Gros Ventre soils. The experiment was carried on for 58 days and the highest rise in unsterilized soil at the end of the experiment was obtained in Fallis soil, the next highest in Edmonton

soil, and the lowest in Gros Ventre. In the sterilized soils the rise was in the same order, although not as high. After 58 days the sterilization had decreased capillary rise by about 49, 62, 53, and 37%, respectively in the Edmonton, Vegreville, Gros Ventre, and Fallis soils.

The differences in capillary rise of water in the different soils, and between sterilized and unsterilized soils, are probably due mainly to differences in texture, quantity, and character of colloidal material present. The quantity of soluble matter, including soluble organic matter, is definitely increased by steaming. Pickering (14) and Lyon and Bizzell (11) found that water soluble matter was increased by steaming.

pH VALUES

In this investigation, as shown in Table II, no significant differences were found in pH values, as measured by the quinhydrone electrode, between sterilized and unsterilized soils after the sterilized soils had been autoclaved

TABLE II
HYDROGEN ION CONCENTRATION OF FOUR STERILIZED AND NON-
STERILIZED ALBERTA SOILS, EXPRESSED AS pH
(Determined by the quinhydrone electrode)

	Sterilized				Non-sterilized			
	1 min.	5 min.	15 min.	30 min.	1 min.	5 min.	15 min.	30 min.
Edmonton	6.0	6.1	6.1	6.0	6.3	6.1	6.2	6.0
black	6.1	6.1	6.1	6.1	6.3	6.1	6.3	6.0
park	6.0	6.1	6.1	6.1	6.2	6.2	6.1	6.2
Average	6.0	6.1	6.1	6.1	6.3	6.1	6.2	6.1
Vegreville	6.3	6.1	6.1	6.2	6.2	6.2	6.1	6.2
black	6.2	6.1	6.1	6.2	6.2	6.1	6.0	6.2
park	6.2	6.1	6.1	6.2	6.3	6.1	6.2	6.1
Average	6.2	6.1	6.1	6.2	6.2	6.1	6.1	6.2
Gros Ventre	5.8	5.7	5.8	5.8	6.0	5.8	5.9	5.9
brown	5.8	5.8	5.8	5.7	5.9	5.8	5.9	5.9
prairie	5.7	5.7	5.8	5.7	5.8	5.8	5.9	5.9
Average	5.8	5.7	5.8	5.7	5.9	5.8	5.9	5.9
Fallis	6.1	6.2	6.2	6.2	6.3	6.1	6.2	6.1
gray	6.1	6.2	6.2	6.2	6.3	6.1	6.2	6.2
wooded	6.1	6.2	6.2	6.2	6.3	6.1	6.2	6.2
Average	6.1	6.2	6.2	6.2	6.3	6.1	6.2	6.2

twice, on successive days, once for three-quarters of an hour and the second time for one-half hour at 120° C. Titratable or total acidity was not determined. Schreiner and Lathrop (17) in 1912 found an increase in titratable acidity when they steam sterilized a soil for three hours at 135° C.

WATER SOLUBLE PHOSPHORUS

Water soluble phosphorus was definitely increased in steam sterilized soils, as shown in Table III. In comparison with unsterilized soil, the quantities were about 2.8 times as great in the Edmonton soil, 1.7 times in the Vegreville,

TABLE III

WATER SOLUBLE PHOSPHORUS IN NON-STERILIZED AND STERILIZED ALBERTA SOILS

(Average p.p.m. phosphorus on water-free basis)

Soil	Non-sterilized			Sterilized		
	No. of determinations	P, p.p.m.	Av. deviation from mean	No. of determinations	P, p.p.m.	Av. deviation from mean
Edmonton black park	12	1.67	.47	18	4.69	.48
Vegreville black park	5	4.17	.22	5	7.01	.20
Gros Ventre brown prairie	4	3.16	.05	4	6.72	.05
Fallis gray wooded	4	1.38	.01	4	1.83	.04
<i>Without ignition</i>						
Vegreville black park	4	2.25	.37	4	3.80	.70
Gros Ventre brown prairie	4	1.00	.05	—	—	—

2.1 times in the Gros Ventre, and there was a slight increase in the Fallis soil. Without the ignition of water extracts to destroy organic matter, the quantities of water-soluble phosphorus were found to be approximately half as great in both unsterilized and sterilized soils.

The increase in water soluble phosphorus content induced by sterilization was large in the soils high in organic matter and relatively small in Fallis soil, which is low in organic matter, suggesting that organic phosphorus compounds are partly broken down or made more soluble and later broken down by ignition.

Schreiner and Lathrop (17) found that steam sterilization increased the solubility of soil constituents, increasing all of the organic constituents that they had isolated from unheated soil, except nucleic acid, which was decomposed.

EASILY SOLUBLE PHOSPHORUS (SOLUBLE AT pH 3)

The quantities of easily soluble phosphorus in the four Alberta soils (1) unsterilized, (2) sterilized, and (3) sterilized and recontaminated with original soil, throughout an incubation period of 20 weeks following sterilization, are shown in Table IV. In general, greater quantities of easily soluble phosphorus were found in the sterilized and sterilized recontaminated black Edmonton and Vegreville soils than in the unsterilized. In the Gros Ventre

TABLE IV
EASILY SOLUBLE PHOSPHORUS (SOLUBLE AT pH 3) IN NON-STERILIZED, STERILIZED,
AND RECONTAMINATED STERILIZED ALBERTA SOILS

(Average of duplicates, p.p.m. phosphorus on water-free basis)

Soil	Treatment	Period of incubation after sterilizing soil and setting up experiment									Final av.
		Right after ster.	Days		Weeks						
			3	6	2	4	8	12	16	20	
			Phosphorus, p.p.m., average of duplicates								
Edmonton black park	Non-sterilized	33	—	32	32	37	46	52	49	55	43
	Sterilized and recontaminated	—	—	40	47	57	52	74	75	80	61
	Sterilized	40	—	42	51	56	56	77	70	63	59
Vegreville black park	Non-sterilized	46	—	82	86	69	96	91	86	108	88
	Sterilized and recontaminated	—	—	78	89	93	125	123	103	113	103
	Sterilized	60	—	83	99	83	103	109	104	110	99
Gros Ventre brown prairie	Non-sterilized	56	48	48	46	56	54	98	56	57	58
	Sterilized and recontaminated	—	48	49	43	53	61	107	61	67	61
	Sterilized	62	56	54	48	52	55	82	53	61	59
Fallis gray wooded	Non-sterilized	87	76	83	82	92	100	115	76	100	90
	Sterilized and recontaminated	—	85	86	82	103	102	120	91	105	97
	Sterilized	99	90	91	82	94	106	125	84	101	97

brown soil, which is low in organic matter, the differences were insignificant, and in the Fallis gray soil, which is also low in organic matter, the differences were probably insignificant, because, while close agreement was obtained in reading the duplicates of a single determination, a variation of 10 p.p.m. was often obtained in separate determinations. In the final averages the increases of sterilized over unsterilized were about 37% in the Edmonton soil, 12% in the Vegreville, 8% in the Fallis, and if anything a little more in the sterilized recontaminated.

Increases in soluble phosphorus found right after sterilization must be attributed to changes brought about by steaming. Increases with time of incubation may be attributed to microbiological activity in the unsterilized

and the sterilized recontaminated soils, following remoistening and cultivation, but this is doubtful because the sterilized soils that were not recontaminated also showed an increase in easily soluble phosphorus.

Darbishire and Russell (7) found that plants grown on partially sterilized soil utilized about twice as much phosphorus as plants grown on normal soil, and that the percentage of phosphorus in plants grown on partially sterilized is much higher than on normal soil. Baldwin (1) found more easily soluble phosphorus in sterilized than in unsterilized soil in the greenhouse.

NITRIFICATION

The results of the two nitrification experiments are shown in Figs. 6, 7, 8, and 9. Greatest nitrification occurred in Vegreville and Edmonton soils, less in Gros Ventre, and least of all in Fallis soil; that is to say, the higher the organic matter content of the soil, the greater the nitrification

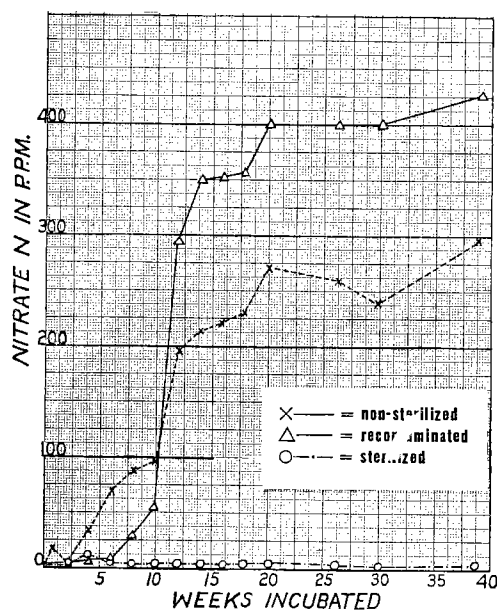


FIG. 6. Nitrification in non-sterilized, recontaminated sterilized, and sterilized Edmonton black park soil.

In all cases the nitrate content remained practically constant and generally very low in the sterilized soils throughout the 39 weeks of incubation. Nitrate accumulated in all of the unsterilized soils. Nitrification was slower at first in the sterilized recontaminated soil but after about 10 weeks, nitrate accumulated more rapidly in the sterilized recontaminated than in the unsterilized. The increase in the rate of nitrification in the sterilized recontaminated is evidently related to the ammonia content of the soil, because the ammonia that had accumulated in these soils began to decrease rapidly in most cases at about 10 weeks or a little earlier.

These results indicate that either the nitrifying organisms (*Nitrosomonas* and *Nitrobacter*) take a long time to become active when re-introduced into

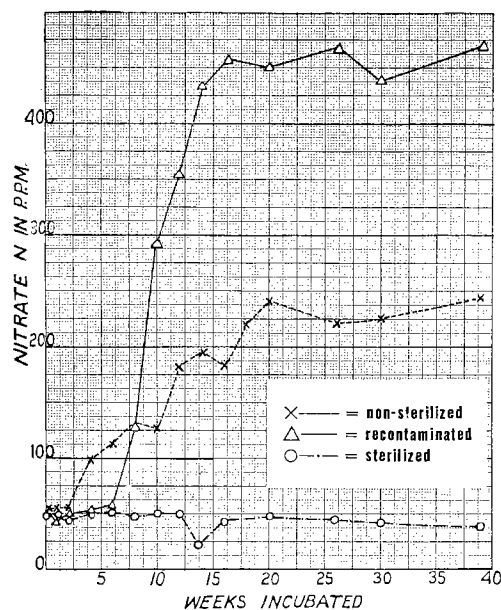


FIG. 7. Nitrification in non-sterilized, recontaminated sterilized, and sterilized Vegreville black park soil.

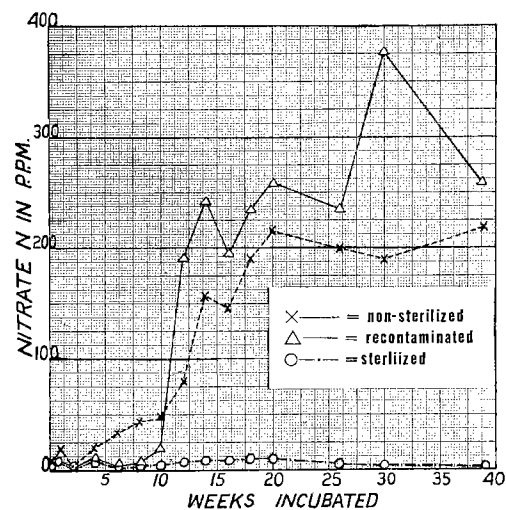


FIG. 8. Nitrification in non-sterilized, recontaminated sterilized, and sterilized Gros Ventre brown prairie soil.

steam sterilized soil, or that sterilization in some way alters the chemical composition of the soil, making it unfavorable for their development until other micro-organisms have had time to overcome the inhibitory factors.

Schreiner and Lathrop (17) found an increase in a large number of organic nitrogen compounds following steam sterilization, and this may account, in part at least, for the greater nitrification in the recontaminated soils. Presumably the same micro-organisms as were present in the unsterilized soil are active, though certain groups may develop more rapidly for a time because of the altered chemical and physical conditions in sterilized soils.

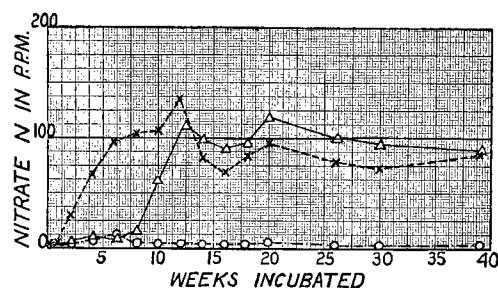


FIG. 9. Nitrification in non-sterilized, recontaminated sterilized, and sterilized Fallis gray wooded soil; X — X = non-sterilized; Δ — Δ = recontaminated; O — O = sterilized.

AMMONIFICATION

The greatest ammonification or accumulation of ammonia occurred in the sterilized soil recontaminated with original soil, as shown in Figs. 10 to 13. In the first four to six weeks of incubation, a steady increase in ammonia took place in three out of four of the recontaminated soils, rising to a high peak,

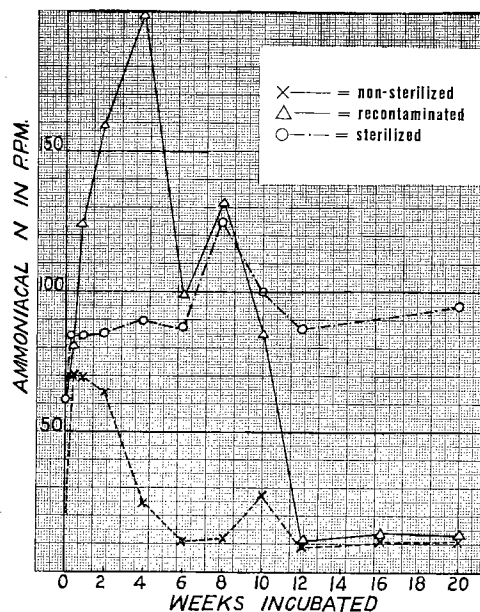


FIG. 10. Ammonification in non-sterilized, recontaminated sterilized, and sterilized Edmonton black park soils.

as may be seen on Figs. 10, 11, and 12. This increase was followed by a rapid return to about the same low level as that of the unsterilized, at about 12

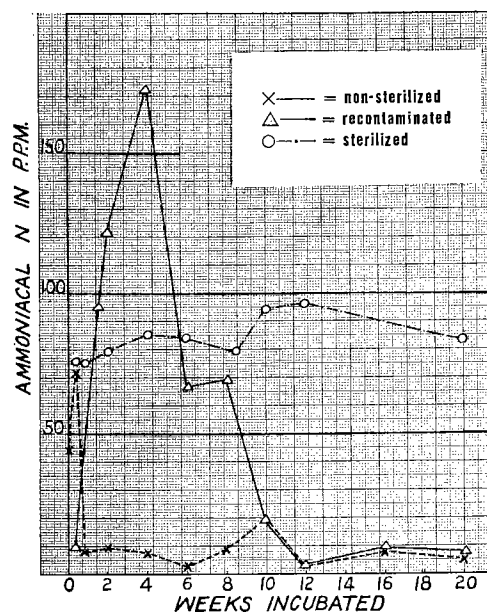


FIG. 11. Ammonification in non-sterilized, recontaminated sterilized, and sterilized Vegreville black park soils.

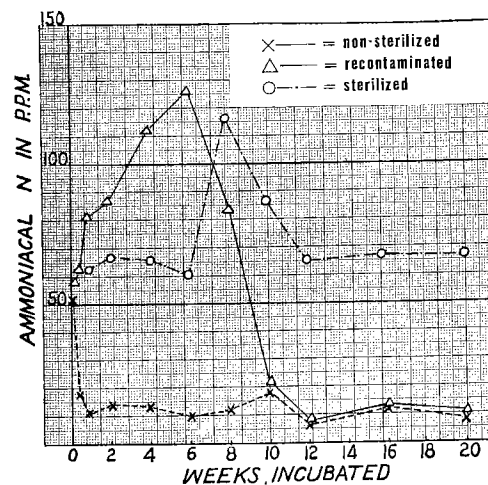


FIG. 12. Ammonification in non-sterilized, recontaminated sterilized, and sterilized Gros Ventre brown prairie soil.

weeks. However, in the case of recontaminated Fallis soil the accumulation of ammonia was relatively small, and it took longer to return to the normal low level. The accumulation of ammonia was roughly proportional to the

organic matter content of the soil: Edmonton, 198 p.p.m. of ammoniacal nitrogen; Vegreville, 172; Gros Ventre, 126; and Fallis, 63 p.p.m.

During the first few days of the experiment there was a temporary increase of ammonia in the unsterilized soils. This was probably the result of bringing

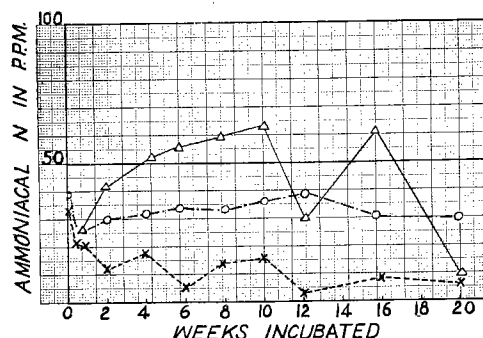


FIG. 13. Ammonification in non-sterilized, recontaminated sterilized, and sterilized Fallis gray wooded soil; X—X = non-sterilized; Δ—Δ = recontaminated; O—O = sterilized.

to optimum conditions of moisture and temperature soils that had been kept air-dry, thus renewing the activity of ammonifying micro-organisms. The nitrifiers apparently do not become active as quickly as the ammonifiers when air-dry soil is remoistened, but after a short period of time they convert nearly all of the available ammonia to nitrate.

The ammonia content of the sterilized soils remained at a fairly constant level above that of the unsterilized. This was probably due to the production of ammonia compounds that were absent before heating (17), but the fluctuations from this more or less constant level for each soil were probably due to contamination from the air, since each flask was opened every week or 10 days in order to add sterilized water. Great care was exercised, when watering flasks, to prevent recontamination, by flaming the mouth of each flask and keeping it open just long enough to add water. Nevertheless the cotton plug through which was placed a stirring rod for cultivating and mixing the soil after watering became loose in some flasks after repeated cultivation.

A shorter experiment on ammonification was also set up as a duplicate of the longer experiment, and ammonia was determined after three and six days, two weeks, and four weeks after sterilization. The results of this experiment are not given because the figures show the same trends as those of the first experiment.

Since the method used to determine ammonia involved the use of magnesium oxide to displace ammonia on distillation of soil extract, it was thought advisable to find out whether magnesium oxide would break down dissolved organic matter and liberate ammonia in stages rather than cause it to be given off completely in one stage. The distillations were carried out under uniform conditions for 45 min. from first boiling, during which time approximately

250 cc. of distillate were collected. When the distillation was prolonged for 45 min., a very small amount of additional ammonia nitrogen was obtained: about 4 p.p.m. as an average of 30 determinations.

For example, in one case more than 90 p.p.m. of ammoniacal nitrogen were distilled off in 10 min., 135 in 15 min., and 180 in 20 min., when the total was less than 200 p.p.m. Distilling for 45 min., from the first boiling, would therefore allow more than enough time for displacement of readily displaceable ammonia.

SULPHOFICATION

Generally speaking there was some increase in water soluble sulphate in all four unsterilized soils on incubation, as shown in Table V, and large increases in all sterilized recontaminated soils, except Fallis. However, the sterilized

TABLE V
SULPHOFICATION IN NON-STERILIZED, STERILIZED, AND RECONTAMINATED
STERILIZED ALBERTA SOILS

(Average of duplicates, p.p.m. sulphate on water-free basis)

Soil	Treatment	Period of incubation after sterilizing soil and setting up experiment				
		Right after ster.	Weeks			
			2	4	8	12
Sulphate, p.p.m., average of duplicates						
Edmonton black park	Non-sterilized	71	64	66	69	79
	Sterilized and recontaminated	98	103	94	126	223
	Sterilized	96	114	79	124	141
Vegreville black park	Non-sterilized	76	68	70	126	101
	Sterilized and recontaminated	136	123	100	158	188
	Sterilized	122	147	85	124	152
Gros Ventre brown prairie	Non-sterilized	42	52	—	89	80
	Sterilized and recontaminated	70	68	86	105	106
	Sterilized	110	102	86	108	126
Fallis gray wooded	Non-sterilized	24	42	81	38	39
	Sterilized and recontaminated	50	42	54	57	52
	Sterilized	43	58	36	47	49

soils also showed some fluctuations and apparent increases in water soluble sulphate on incubation, and the results as a whole are not entirely consistent.

Sterilization alone considerably increased the sulphate content of each soil, as shown in the first column of Table V, but this may be due merely to an increase in soluble organic matter, which on ignition gives higher sulphate

TABLE VI
WATER SOLUBLE SULPHATES WITH AND WITHOUT IGNITION
(Average of duplicates, p.p.m. SO₄ on water-free basis)

Soil (incubated 12 weeks)	Treatment	Without ignition, SO ₄ p.p.m.	With ignition, SO ₄ p.p.m.
Edmonton black parkland	Non-sterilized	49	79
	Sterilized and recontaminated	185	222
	Sterilized	81	140
Vegreville black parkland	Non-sterilized	64	100
	Sterilized and recontaminated	127	187
	Sterilized	95	151
Gros Ventre brown prairie	Non-sterilized	45	79
	Sterilized and recontaminated	83	105
	Sterilized	70	125
Fallis gray wooded	Non-sterilized	18	38
	Sterilized and recontaminated	23	51
	Sterilized	15	48

results. Increases in sulphate on incubation are probably brought about by micro-organisms that oxidize proteins and other forms of organic matter that contain sulphur.

A comparison of the amounts of water soluble sulphate found with and without ignition of evaporated extracts is shown in Table VI. Without ignition only about two-thirds as much sulphate was obtained as with ignition, indicating that part of the sulphate obtained on ignition comes from soluble organic matter.

Acknowledgments

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References

- BALDWIN, I. L. Greenhouse soil sterilization. II. Bacteriological and chemical examination. Indiana Agr. Expt. Sta. Bull. 266. 1922.
- BEWLEY, W. F. Practical soil sterilization. Ministry of Agriculture and Fisheries. Bull. 22. England. 1935.
- BOUYOUCOS, G. J. Directions for making mechanical analyses of soils by the hydrometer method. Soil Sci. 42 : 225-228. 1936.
- BROWN, H. D. Greenhouse soil sterilization. I. Sterilization methods and pest control. Indiana Agr. Expt. Sta. Bull. 266. 1922.
- COLEMAN, D. A. Environmental factors influencing the activity of soil fungi. Soil Sci. 2 : 1-65. 1916.
- CONNER, S. D. Greenhouse soil sterilization. III. Excess soluble salts in greenhouse soils. Indiana Agr. Expt. Sta. Bull. 266. 1922.
- DARBISHIRE, F. V. and RUSSELL, E. J. Oxidation in soils and its relation to productivity. Part II. The influence of partial sterilisation. J. Agr. Sci. 2 : 305-326. 1907.

8. HAINES, W. B. The volume-changes associated with variations of water content in soil. *J. Agr. Sci.* 13 : 296-310. 1923.
9. HARPER, H. J. The accurate determination of nitrates in soils. Phenoldisulfonic acid method. *J. Ind. Eng. Chem.* 16 : 180-183. 1924.
10. LYON, T. L. and BIZZELL, J. A. Effect of steam sterilization on the water soluble matter in soils. *Cornell Agr. Expt. Sta. Bull.* 275. 1910.
11. LYON, T. L. and BIZZELL, J. A. Water-soluble matter in soils sterilized and reinoculated. *Cornell Agr. Expt. Sta. Bull.* 326. 1913.
12. MCLEAN, W. and ROBINSON, G. W. A new method for the determination of ammoniacal nitrogen in soils. *J. Agr. Sci.* 14 : 548-554. 1924.
13. NEWHALL, A. G. Theory and practice of soil sterilization. *Agr. Eng.* 16 : 65-70. 1935.
14. PICKERING, S. U. Studies of the changes occurring in heated soils. *J. Agr. Sci.* 3 : 258-284. 1910.
15. PARKER, F. W. and FUDGE, J. F. Soil phosphorus studies. I. The colorimetric determination of organic and inorganic phosphorus in soil extracts and the soil solution. *Soil Sci.* 24 : 109-117. 1927.
16. RUSSELL, E. J. and HUTCHINSON, H. B. The effect of partial sterilisation on the production of plant food. *J. Agr. Sci.* 3 : 111-144. 1909.
17. SCHREINER, O. and LATHROP, E. C. The chemistry of steam heated soils. *U.S.D.A. Bur. Soils Bull.* 89. 1912.
18. SENNER, A. H. Application of steam in the sterilization of soils. *U.S.D.A. Tech. Bull.* 443. Washington, D.C. 1934.
19. TRUOG, E. The determination of the readily available phosphorus of soils. *J. Am. Soc. Agron.* 22 : 874-882. 1930.